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2 **Synergistic effect of MoS₂ and diamond nanoparticles in**
3 **electrochemical sensors : determination of an anticonvulsant drug**

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Abstract

We have developed an electrochemical sensor based on the employment of two new emerging nanomaterials: diamond nanoparticles (DNP) and molybdenum disulfide (MoS_2). The sensor was applied to the determination of an anticonvulsant, valproic acid, previously derivatized with a ferrocene group. MoS_2 platelets were obtained by an exfoliation method and DNP were directly dispersed in water and subsequently employed for glassy carbon (GC) electrodes modification. The sensor response was optimized in terms of both the solvent employed for dispersing the MoS_2 nanomaterial and the method for GC electrode modification. The better response was obtained for sensors based on a first layer of MoS_2 dispersed in ethanol:water and a second DNP layer. The different steps of the sensor construction were characterized by atomic force microscopy (AFM) and electrochemical impedance spectroscopy (EIS). In order to evaluate the synergetic effect of DNP and MoS_2 , the differential pulse voltammetry (DPV) sensor response (measured at +0.18 V) was compared with that obtained, at the same potential, for sensors incorporating only one of the nanomaterials (DNP or MoS_2). Results demonstrated that the formation of a hybrid MoS_2 -DNP structure clearly enhances the performance of the sensor. The sensor containing both nanomaterials exhibits a response with high sensitivity (0.74 A M^{-1}), and good detection limit and reproducibility ($0.27 \text{ }\mu\text{M}$ and 8%, respectively). Moreover, after 45 days the sensor retains the 99% of the initial response, showing excellent storage stability.

Keywords: molybdenum disulfide, diamond nanoparticles, valproic acid, electrochemical sensor, atomic force microscopy (AFM), electrochemical impedance spectroscopy (EIS), differential pulse voltammetry (DPV).

Introduction

Molybdenum disulfide, a member of the transition metal dichalcogenides (TMDs) family, has attracted a great interest. These emerging 2D nanomaterials, analogous to graphene, present unique mechanical, electrical, thermal and optical properties, opening up a challenging new research field [1-3].

MoS₂ nanosheets result from stacking units, formed by three atomic sulfur-molybdenum-sulfur (S-Mo-S) layers, through van der Waals interactions. In this sandwich structure, Mo atoms are coordinated in a trigonal prismatic geometry to six S atoms. There are different methods to obtain MoS₂ nanosheets based on top-down or bottom-up approaches [2]. Top-down strategies consist in breaking down bulk MoS₂ into thin layers, for example by liquid exfoliation [4-6] or by intercalating ions [7] into the weakly stacked layers in order to expand them. Conversely, the bottom-up strategies lie on the synthesis of MoS₂ from building blocks such as molybdenum metal salts and organic compounds containing sulfur [8,9]. One of the most commonly employed bottom-up technique is the chemical vapor deposition [10].

MoS₂ has found its main applications in optoelectronics, generation and storage of energy i.e., supercapacitors and batteries [11-14]. In addition, the possibility of employing this nanomaterial for chemical sensing and biosensing is also attracting an increasing attention [15-19]. However, efforts devoted to employ this nanomaterial for sensing, has revealed some drawbacks. In particular, for electrochemical sensor applications, the inert nature of the basal planes, the limited number of edge sites and the low conductivity of MoS₂ nanosheets represent important disadvantages [20]. To overcome limitations and improve the activity of the MoS₂ nanosheets, an interesting strategy is related to generate hybrid structures formed by MoS₂ and conductive materials [15,16,20]. Among others, hybrids formed by MoS₂ and metal nanoparticles

[21,22], graphene [23,24], carbon nanotubes [25] or conducting polymers [26-28] have been developed and employed as sensing platforms to detect several compounds such as eugenol, glucose and dopamine [29-31].

On the other hand, diamond nanoparticles (DNP), has also gained attention as a promising nanomaterial for electrochemical sensing applications [32-34]. In particular, it presents several advantages compared to other carbon nanomaterials such as its noncytotoxic nature, good biocompatibility and low-cost.

Keeping in mind the excellent individual properties of both nanomaterials, the aim of the present work is to prove that their combination can offer new opportunities to develop electrochemical sensors with improved performances. As a proof of concept, we have evaluated the potential synergetic effect of MoS₂/DNP for the determination of valproic acid, previously derivatized with a ferrocene group (VA-Fc), to render it electroactive.

Valproic acid (VA) is an anticonvulsant employed to treat some disorders such as epilepsy, bipolar disorders and migraine headaches. Drug monitoring is important since VA displays therapeutic efficiency for plasma concentrations in the range 50-100 ppm, but can induce toxicity for higher concentrations. Analytical techniques reported in the literature for VA determination include high liquid performance chromatography (HPLC) with different detection systems, gas chromatography coupled with mass spectrometry (GC-MS), capillary electrophoresis and immunological assay, among others [35-47]. Most of these techniques usually involve a previous derivatization step in order to render the analyte suitable for detection. For example, techniques employing UV or fluorescence detection require derivatization, due to the absence of a strong chromophore or fluorophore in VA molecule. In the case of electrochemical detection, derivatization with a suitable redox group is also needed.

In the present work, we have modified for the first time a GC electrode with MoS₂ and DNP in order to develop an electrochemical sensor. The resulting device has been characterized by atomic force microscopy (AFM) and electrochemical impedance spectroscopy (EIS). AFM allows us to determine the lateral dimensions and thickness of the MoS₂ sheets, as well as the morphology of the sensing surface (MoS₂/DNP). EIS measurements allow evaluating the resistance charge transfer (R_{CT}) of MoS₂, DNP and MoS₂/DNP modified GC surfaces. The applicability of the sensor was tested for valproic acid determination (previously derivatized with a ferrocene group) in real serum samples.

Experimental Section

Materials

Diamond nanoparticles (DNP) were obtained from SkySpring Nanomaterials (Product 0512HZ, nominal diameter 4-15 nm, www.ssnano.com), Inc (Houston, Tx). Molybdenum disulfide (99%), N-methyl pyrrolidone (NMP), dimethylformamide (DMF), ethanol (EtOH), isopropyl alcohol (IPA), methanol, dichloromethane, triethylamine, dopamine, ascorbic acid, epinephrine, lithium tetrahydridoaluminate, potassium ferrocyanide, potassium ferricyanide, potassium cyanide, and potassium chloride were obtained from Aldrich Chemical Co (www.sigmaaldrich.com). 2,2-di-n-propylacetic acid and *N,N,N*-trimethylferrocenylmethyl-amonium iodide were obtained from Alfa Aesar (www.alfa.com/es). Sodium phosphate (www.merck.com) was employed for the preparation of buffers. Water was purified with a Millipore Milli-Q-System (www.millipore.com). All solutions were prepared just prior to use.

Experimental techniques

The AFM measurements were performed in air with Nanoscope IIIa (Veeco, www.veeco.com) and Agilent 5500 (Agilent, www.agilent.com) systems. The images were taken in the dynamic mode using silicon cantilevers (Bruker, www.bruker.com) with a nominal force constant of 40 N/m and a nominal radius of 8 nm. First, large areas (around 100 μm^2) were scanned in order to locate the MoS₂ flakes, which were then imaged at higher resolution. The images, 1024 x 1024 pixels, were taken with different cantilevers in order to ensure that the imaged structures were not due to tip artefacts. Also, eventually imaging under liquid conditions was performed in order to discard artefacts coming from remaining water used in the sample preparation. Supports used were Si substrates from University Wafer (www.universitywafer.com, USA).

Impedance and voltammetric studies were carried out with an Ecochemie Autolab PGSTAT302 N system (Utrecht, The Netherlands, www.ecochemie.nl) in a cell with a working GC electrode (3 mm internal diameter) and a platinum wire as counter electrode (www.metrohm.com). All potentials were reported with respect to a Ag/AgCl reference electrode. DPV measurements were carried out in a 0.1 M phosphate buffer (pH = 7) containing VA-Fc, with the following conditions: range potential from -50 mV to 450 mV, scan rate 10 mVs⁻¹, pulse amplitude 60 mV and step potential 1 mV. EIS experiments were performed in a 0.1 M phosphate buffer (pH = 7), containing 10 mM K₃Fe(CN)₆/ K₄Fe(CN)₆. A sinusoidal potential modulation of ± 10 mV amplitude in the 10⁵ Hz-10⁻² Hz frequency range, spaced logarithmically (120 per 8 decades), was superimposed onto the formal potential of the redox couple, [Fe(CN)₆]^{3-/4-}.

HPLC measurements were performed with a Jasco Analytica PU-1580 high pressure pumping system, equipped with Rheodyne Model 7125 injector with a 20 μL loop and a

Kromasil C18 column (150 x 4.6 mm; 5 μ m; Scharlau). A Perkin Elmer 785A UV/VIS was employed as detector at λ = 260 nm.

Functionalization of valproic acid with aminoethyl-substituted ferrocene

Organometallic compound $\text{Fe}\{[(\eta^5\text{-C}_5\text{H}_4)(\text{CH}_2)_2\text{NHC(O)CH}(\text{CH}_2\text{CH}_2\text{CH}_3)_2](\eta^5\text{-C}_5\text{H}_5)\}$ (VA-Fc) was synthesized by direct amide coupling of equimolar amounts of β -aminoethylferrocene $\text{Fe}\{(\eta^5\text{-C}_5\text{H}_4(\text{CH}_2)_2\text{NH}_2)(\eta^5\text{-C}_5\text{H}_5)\}$ (500 mg, 2.18 mmol) with 2,2-di-*n*-propylacetic acid, in dry CH_2Cl_2 , in the presence of triethylamine. The solution was concentrated and the solid formed was separated by filtration and dried under vacuum to afford the desired compound VA-Fc which was isolated as a yellow-orange solid (yield: 53%). Data obtained from ^1H NMR, IR and MS (FB^+) confirm that VA-Fc has been successfully synthesized (data not shown).

Compound $\text{Fe}\{(\eta^5\text{-C}_5\text{H}_4(\text{CH}_2)_2\text{NH}_2)(\eta^5\text{-C}_5\text{H}_5)\}$ [48, 49] was selected as starting metallocene since the reactive NH_2 functionality is separated from the cyclopentadienyl ring by a two methylene flexible bridge. This fact is of importance because it minimizes steric and electronic effects of the organometallic ferrocene moiety. The use of a β -functionalized metallocene, in addition, avoids the instability of α -functional ferrocene derivatives resulting from the stability of the α -ferrocenyl carbonium ion.

Concerning monofunctionalized ferrocene $\text{Fe}\{(\eta^5\text{-C}_5\text{H}_4(\text{CH}_2)_2\text{NH}_2)(\eta^5\text{-C}_5\text{H}_5)\}$ this compound was prepared in two steps from *N,N,N*-trimethylferrocenylmethyl-amonium iodide as starting material by adapting the literature procedures [50]. Firstly, $[(\eta^5\text{-C}_5\text{H}_5)\text{Fe}\{\eta^5\text{-C}_5\text{H}_4\}\text{CH}_2\text{N}(\text{CH}_3)_3]\text{I}$ was reacted with potassium cyanide thus resulting 1-cyanomethylferrocene which was isolated as a yellow crystalline solid in 63% yield. The cyanomethylferrocene was converted into β -aminoethylferrocene by reduction with LiAlH_4 followed by treatment with aqueous sodium hydroxide. After aqueous workup

and distillation in vacuum, primary amine $\text{Fe}\{(\eta^5\text{-C}_5\text{H}_4(\text{CH}_2)_2\text{NH}_2)(\eta^5\text{-C}_5\text{H}_5)\}$ was obtained as an amber-brown oil in 66% yield.

VA-Fc was dissolved in methanol, giving rise to a 2.2×10^{-3} M VA-Fc stock solution. Working solutions were prepared by dilution of the stock solution in phosphate buffer 0.1 M pH=7.0.

Synthesis of MoS₂ platelets

MoS₂ nano-sized powders were employed as starting materials to obtain MoS₂ atomic layers following a liquid exfoliation strategy [4], with some modifications. The solvents selected as exfoliation mediums were NMP, DMF, EtOH/water (45:55) and IPA/water (45:55). Briefly, 75 mg of the commercial MoS₂ powder was mixed with 10.0 mL of the corresponding solvent. After sonication of each dispersion for 2 hours, it was kept at 4 °C during 24 hours and then centrifuged (1500 rpm, 45 minutes). The supernatant was collected and subsequently employed for electrode modification.

Preparation of the samples for AFM measurements

Samples for AFM measurements were prepared by modifying Si surfaces as follows: i) placing 6 μL of MoS₂ exfoliated in EtOH/water, ii) leaving air-dry and iii) depositing 6 μL of DNP dispersed in water (1 mg mL^{-1}). An additional sample was prepared in the same way but employing DNP diluted 1:1000 with water ($1 \mu\text{g mL}^{-1}$).

Preparation of the electrochemical sensors. Evaluation of the response.

GC electrodes were polished with 1 μm diamond paste (Buehler) and rinsed with water. Subsequently, four different sensors were obtained by modifying these GC electrodes as follows: i) placing 6 μL of MoS₂ exfoliated in NMP, DMF, EtOH/water or

IPA/water solvent, ii) leaving air-dry, iii) depositing 6 μL of DNP dispersed in water (1 mg mL^{-1}) and air-dry again. For comparison, four other electrochemical sensors consisting in a first layer of DNP and a second layer of MoS_2 (exfoliated in NMP, DMF, EtOH/water or IPA/water) were also prepared following the same procedure but inverting the order of MoS_2 and DNP deposition.

Sensors containing only MoS_2 or DNP were obtained by modifying the GC electrode with 6 μL of the corresponding nanomaterial and leaving air-dry it.

The sensor response was evaluated as follows: the electrodes, modified as described above, were introduced into an electrochemical cell containing 3.5 μM of VA-Fc in 0.1 M phosphate buffer (pH = 7) and the differential pulse voltammograms were obtained (see DPV conditions in the experimental techniques section). The calibration curve was obtained: i) recording DPVs for different VA-Fc concentrations in the range potential from -50 mV to 450 mV, ii) plotting the intensity current obtained at +0.18 V towards the corresponding VA-Fc concentration.

Evaluation of the method: VA-Fc analysis in serum samples.

Human blood was centrifuged at 4 $^{\circ}\text{C}$ for 10 min at 2500 rpm. The serum was then separated with a syringe and stored at 4 $^{\circ}\text{C}$ prior to analysis. Samples were diluted 1:100 with phosphate buffer (pH 7) and spiked with five different VA-Fc concentrations (from $1.4 \cdot 10^{-6}$ M to $5.5 \cdot 10^{-6}$ M). The determination was carried out by the DPV method in order to calculate the recoveries.

HPLC-UV measurements were performed for comparison. Chromatograms were obtained at a flow rate of 1.0 mL min^{-1} , employing EtOH/ NaH_2PO_4 0.01 M (60:40, v/v) as mobile phase, pH 6. A wavelength of 260 nm was employed for detection. Serum samples were diluted 1:100 in mobile phase and spiked to obtain final VA-Fc

concentrations of 1.0×10^{-5} , 2.5×10^{-5} and 5.0×10^{-5} M. Samples were filtered through a 0.45 μm pore size nylon filter.

Results and discussion

We have developed an electrochemical sensor by combining two nanomaterials, DNP and MoS_2 , which are attracting an emerging attention in the sensors field. As mentioned in the introduction, several (bio)sensors have been developed using one of these two nanomaterials, leading to improved performances. Thus, their excellent individual properties suggest the possibility of giving rise to a synergistic effect when employed together. In particular, the presence of DNP can overcome one of the main drawbacks of employing MoS_2 in electrochemical sensors, related to its low conductivity. Until now, hybrids formed by MoS_2 and different nanomaterials such as graphene or carbon nanotubes have been assayed as sensor platforms, but DNP have not been tested for this goal.

Optimization of the sensor construction

Following the procedure described in the experimental section, we have exfoliated molybdenum disulfide by sonication in four different solvents to produce stable MoS_2 dispersions sheets. In order to evaluate the suitability of a given solvent, we have obtained the response of four different sensors fabricated by modifying GC electrodes with DNP and MoS_2 previously exfoliated in NMP, DMF, EtOH/ H_2O and IPA/ H_2O , respectively. The differential pulse voltammograms, displayed in Figure 1, are obtained in a 0.1 M phosphate buffer pH=7.0, containing 3.5 μM of VA-Fc. Although a well-defined oxidation peak, corresponding to VA-Fc, is observed for all cases, the response is clearly better when EtOH/ H_2O is employed as solvent. In particular, the peak current

measured at a potential of +0.18 V is approximately three times higher in comparison to the current obtained for the rest of solvents.

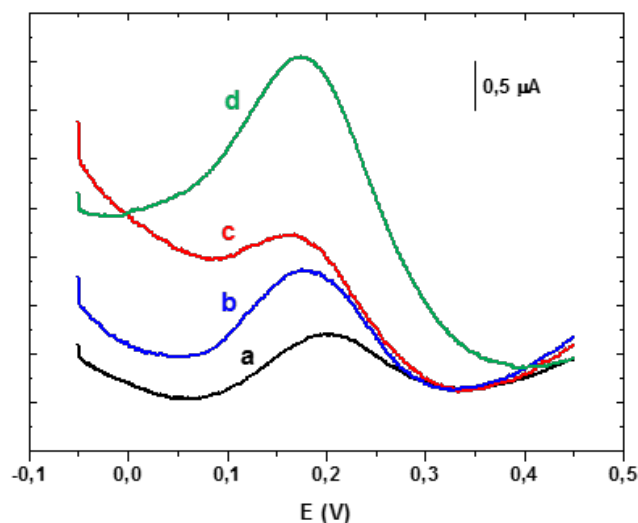


Figure 1. DPV response of GC electrodes modified with DNP and MoS₂ in a 0.1 M phosphate buffer pH=7.0, containing 3.5 μM of VA-Fc. MoS₂ employed for the sensors fabrication had been previously exfoliated employing as solvent: (a) NMP, (b) DMF, (c) IPA/water and (d) EtOH/water. DPV conditions: range potential from -50 mV to 450 mV, scan rate 10 mV s⁻¹, pulse amplitude 60 mV and step potential 1 mV.

Once EtOH/H₂O was selected as solvent, we proceeded to optimize the modification procedure. In order to do this, two different sensor configurations have been compared. The first one corresponds to that described until now, a first layer of MoS₂ in direct contact with the GC surface, and a second layer of DNP in contact with the medium. The response of this sensor (GC/MoS₂/DNP) has been previously shown (Figure 1, curve d). The second configuration is just the opposite, an inner DNP layer in contact with the electrode and an outer MoS₂ layer. The response obtained with this configuration (GC/DNP/MoS₂) is similar or even higher (data not shown), than that corresponding to GC/MoS₂/DNP. However, this sensor arrangement was discarded due to its poor stability. In particular, the DPV measured after 3 days is similar to that obtained for the GC/DNP system, suggesting that the MoS₂ layer is partially lost when

it is directly exposed to the medium. When this layer is protected by DNP the stability sensor is highly improved as it will be discussed later.

Finally, some experimental conditions, such as pH, electrolyte and pulse amplitude in DPV measurements were optimized. In particular, the response of a GC/MoS₂/DNP electrode was obtained in the following electrolytes, containing 3.5 μ M of VA-Fc: acetate buffer at pH=4.0, phosphate buffer at pH=7.0 and borate buffer at pH=10.0. Concerning pulse amplitude, values in the range 10-80 mV were assayed. The best results were obtained for phosphate buffer at pH 7.0 and a pulse amplitude of 60 mV.

Morphological sensor characterization: AFM measurements

The different steps of the sensor construction were characterized by atomic force microscopy (AFM). Figure 2A shows a 5 x 5 μ m² image in which several MoS₂ platelets can be observed. They have a height in the 4-6 nm range. Their size varies from few nm² to areas above 1 μ m². This sample in fact corresponds to a MoS₂ sample on which DNP (from a diluted stock, 1 μ g mL⁻¹) were deposited (see the experimental section). It can be distinguished how these DNP are deposited, mainly as aggregates, on top of the large MoS₂ platelets. When the background of the image is inspected in detail, a small structure can be discerned. When this zone is imaged at higher resolution, images as that of Figure 2B are obtained. Here, small plateaus are observed with relatively jagged perimeters and lateral sizes below 200 nm. From the height distribution of the image (Figure 2C) it is clearly obtained that the average measured height of these flakes is close to 0.8 nm, i.e., very close (within the error bar) to that of a MoS₂ monolayer. It should be noted that, in this height histogram, we have taken the zero height value at the lowest point of the whole image and not at the average height of the flat background (which is located in the histogram at 0.6 nm). Thus, the MoS₂

thickness is the height difference, along the x-axis of the histogram, between both peaks. In Figure 2D is displayed an image taken on a sample prepared by using the same concentration of both the DNP and MoS₂ employed in the samples characterized electrochemically. In particular, now, the DNP number is considerably higher than in Figure 2A since a DNP stock of 1 mg mL⁻¹ was employed for modification. Notwithstanding, once more rather flat platelets are imaged with heights in the 3-5 nm range. However, most of them are almost totally covered by DNP (for instance the large one at the top left corner of the image) while others (as that at the top right corner and the one close to the center of the image) have only some DNP on their surface. The DNP height is in the 5-15 nm range. Interestingly, they tend to aggregate on these partially DNP-covered MoS₂ flakes mainly at their perimeter, i.e., at the steps. In contrast with Figure 2A, now the DNP are also found on the substrate, outside the MoS₂ flakes, which is likely due to their high amount. Finally, in Figure 2E is displayed a zoom of one of these partially DNP-covered MoS₂ flakes. This flake has a thickness close to 5.5 nm and the DNP on top are 8-15 nm high. At given spots, mostly at the perimeter of the flake, aggregation of DNP is observed.

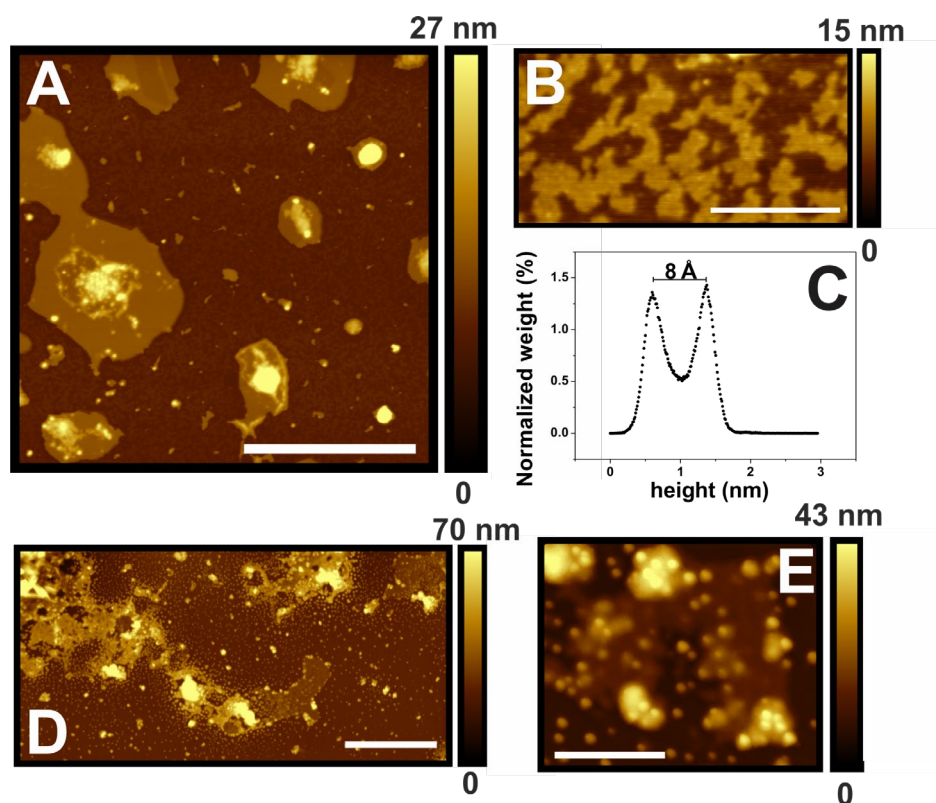


Figure 2. (A) AFM image of SiO₂ substrates modified with MoS₂ and DNP for a low DNP concentration. (B) High resolution image of the flat background revealing the presence of small MoS₂ flakes. Its corresponding height distribution is shown in (C). Note that we have taken the zero height value at the lowest point of the whole image. (D) AFM image of SiO₂ substrates modified with MoS₂ and DNP nanostructures for a high DNP concentration. (E) Detail of a MoS₂ flake partially covered by DNP. The horizontal bars indicate (A) 2 μ m, (B) 500 nm, (D) 3 μ m and (E) 500 nm, respectively. Note that (B) and (E) are not zoomed regions of (A) and (D), respectively but rather higher resolution images taken on similar zones of the sample.

Evaluation of the MoS₂ and DNP synergetic effect in electrochemical sensing

In order to evaluate the synergetic effect of both nanomaterials, the response of the GC/MoS₂/DNP sensor was compared to that of sensors containing only one of the nanostructures, DNP or MoS₂. Figure 3 shows the differential pulse voltammograms of bare GC, GC/MoS₂, GC/DNP and GC/MoS₂/DNP systems. The oxidation peak, corresponding to VA-Fc, is observed for all cases, but its intensity current (measured at +0.18 V) depends on the specific electrode modification. Bare GC electrodes (curve a) and MoS₂-based sensors (curve b) lead to a similar intensity current response, lower than that corresponding to DNP-based ones (curve c). As displayed in curve d, the

simultaneous presence of MoS₂ and DNP gives rise to an intensity current response that is nine and five times higher than that of GC/MoS₂ and GC/DNP, respectively. This result indicates that employing MoS₂ and DNP together as sensing elements allows the development of electrochemical sensors with an enhanced performance. Although we have only tested it in the particular case of VA-Fc determination, this promising starting point suggests the possibility of a broad applicability for determination of other analytes.

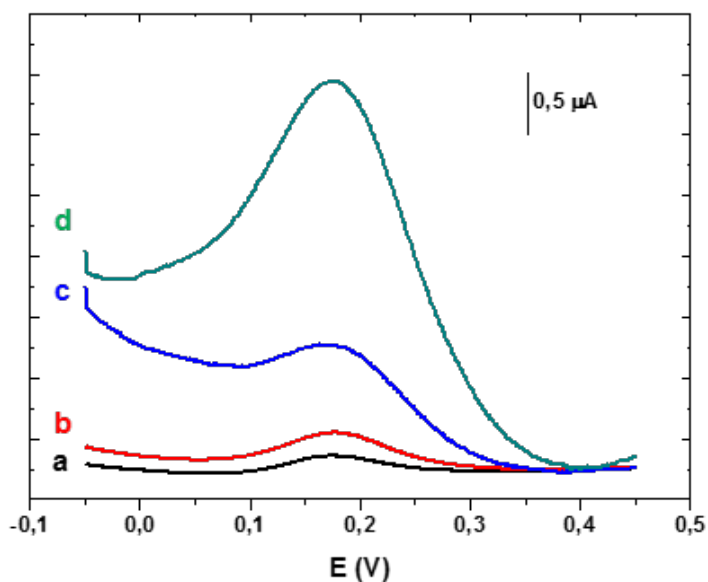


Figure 3. DPV response of (a) bare GC, (b) GC/MoS₂, (c) GC/DNP and (d) GC/MoS₂/DNP electrodes in a 0.1 M phosphate buffer pH=7.0, containing 3.5 μM of VA-Fc. MoS₂ employed for the sensor fabrication has been previously exfoliated employing as solvent EtOH/water. DPV conditions: range potential from -50 mV to 450 mV, scan rate 10 mV s⁻¹, pulse amplitude 60 mV and step potential 1 mV.

The improved response obtained for the sensor containing both nanomaterials is likely due to an increase of both the effective surface area of the electrode and the efficiency in the charge transfer caused by the concomitant presence of MoS₂ and DNP. In order to confirm the latter point, the interfacial properties of bare and modified glassy carbon electrodes were studied by EIS. Figure 4 displays the Nyquist diagrams for GC,

GC/MoS₂, GC/DNP and GC/MoS₂/DNP systems. The semicircles correspond to the charge transfer resistance (R_{CT}) limiting process that is associated with the surface/electrolyte interface. The linear part corresponds to the diffusion-controlled process.

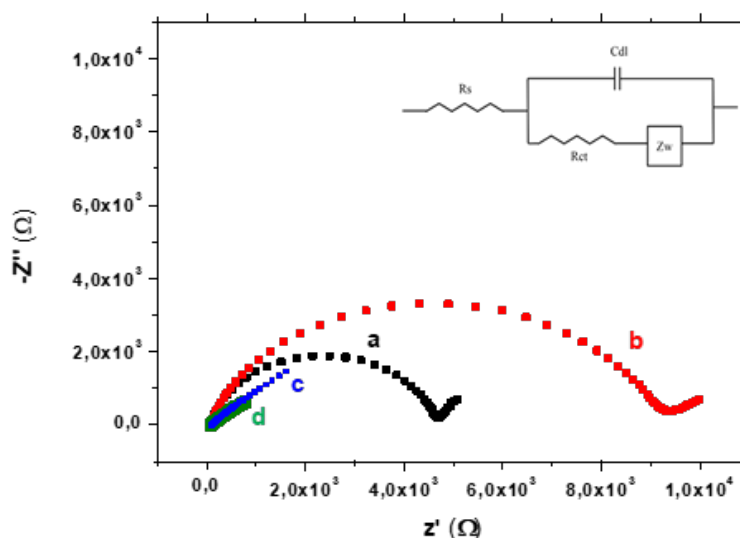


Figure 4. Nyquist plots of (a) bare GC, (b) GC/MoS₂, (c) GC/DNP and (d) GC/MoS₂/DNP electrodes obtained in a 0.1 M phosphate buffer (pH = 7) containing 10 mM [Fe (CN)₆]^{3-/4-} by applying a sinusoidal potential modulation with amplitude of ± 10 mV over the frequency range 10^5 Hz- 10^{-2} Hz. Inset: Randles electrical equivalent circuit.

The impedance diagrams were fitted considering the Randles electrical equivalent circuit, shown in the inset of Figure 4, where R_s , R_{CT} , C_{dl} and Z_w are the electrolyte resistance, the charge transfer resistance, the double layer capacitance and the Warburg impedance, respectively. From the fit, R_{CT} values of 4536.4 Ω and 9000 Ω were obtained for GC and GC/MoS₂, respectively. Upon modification of the GC surface with MoS₂ (curve b), it results an increase of the charge transfer resistance value with respect to the bare GC electrode (curve a). This fact suggests that MoS₂ hinders the electron transfer between the redox probe, [Fe (CN)₆]^{3-/4-} and the glassy carbon electrode. In contrast, for both the GC/DNP and the GC/MoS₂/DNP systems, just a straight line with

a slope close to 1 (0.94 and 0.93, respectively) is obtained (curves c and d). Therefore, the presence of DNP in the surface clearly improves the charge transfer, being a diffusion-limited transport process observed. This behavior confirms that MoS₂/DNP hybrid structures improve the electron transfer with respect to a MoS₂ modified surface, rendering it very attractive for electrochemical sensors development.

On the other hand, the electrochemical surface area was determined by recording five consecutive cyclic voltamograms at 50 mVs⁻¹ with the different modified electrodes in a KCl 1M solution containing 5.0 mM K₄[Fe(CN)₆] / K₃[Fe(CN)₆]. The mean measured peak current (I_p) is related with the electrochemical surface area (A) through the Randles-Sevick equation:

$$I_{pa} = (2.69 \times 10^5) n^{3/2} A D^{1/2} v^{1/2} C$$

where v is the scan rate, n the number of electrons, D the diffusion coefficient (6.5 10⁻⁶ cm² s⁻¹ at 20 °C [51]) and C the concentration of the redox compound. Electrochemical surface areas of (0.065 ± 0.004) cm², (0.0784 ± 0.0005) cm² and (0.0896 ± 0.0003) cm² were obtained for GC, GC/DNP and GC/MoS₂/DNP electrodes, respectively. These values confirm that the presence of both nanomaterials results in a considerable increase of the effective area.

Response of the GC/MoS₂/DNP sensor towards increasing VA-Fc concentration

Once the synergetic effect of MoS₂ and DNP has been assessed, DPVs of GC/MoS₂/DNP at increasing VA-Fc concentration were recorded (Figure S1). The current measured at a potential of +0.18 V was plotted as a function of the VA-Fc concentration in solution (data not shown), being linear in the range from 9.0 10⁻⁷ M to 5.5 10⁻⁶ M, according to the following equation $I (A) = 7 \cdot 10^{-8} (\pm 3 \cdot 10^{-8}) + 0.74 (\pm 0.01) C (M)$, with a correlation coefficient of 0.990. The sensitivity, calculated as the slope of

the calibration curve, was 0.74 A M^{-1} ($8.26 \text{ A M}^{-1} \text{ cm}^{-2}$, when normalized to the electrochemical area). The detection and quantification limits, obtained as the ratio between three and ten times the standard deviation of the blank signal and the sensitivity, were $2.7 \cdot 10^{-7} \text{ M}$ and $9.0 \cdot 10^{-7} \text{ M}$, respectively. As can be observed in Table S1, the detection limit is similar or even better than those obtained by others methods commonly employed for VA determination [35-47]. Note that some of the analytical techniques reported in the literature require the derivatization of VA. For example, for optical detection, derivatization with a chromophore/fluorophore is usually performed in an attempt of improving sensitivity, given the poor valproic acid UV absorption capacity. In our case, for electrochemical detection, derivatization with a suitable redox group is also needed, in order to obtain a redox response at around $+0.2 \text{ V}$, which minimize the potential interferences. From the point of view of derivatization, our method does not represent an improvement with respect to the other techniques usually employed, such as HPLC or GC-MS, but it is less expensive and less time consuming, requiring also less sophisticated equipments.

Finally, reproducibility and stability were evaluated. Reproducibility was estimated by measuring $2.2 \cdot 10^{-6} \text{ M}$ of VA-Fc with parallel GC/MoS₂/DNP sensors. A relative standard deviation value (RSD) of 8% is obtained for five different measurements ($n=5$). Concerning operational stability, a RSD value of 6% was obtained for 50 consecutive measurements with the same GC/MoS₂/DNP sensor. Storage stability was also studied by recording DPVs of a GC/MoS₂/DNP sensor every 15 days. The sensor was kept at $4 \text{ }^{\circ}\text{C}$ between measurements. Even after 45 days, 99 % of the original response was retained, indicating an excellent stability.

In order to evaluate the selectivity, the influence of several potential interferences on the sensor response was studied. In this sense, we have obtained the sensor response

before and after adding increasing amounts of dopamine, ascorbic acid and epinephrine to a solution containing 1.0×10^{-6} M of VA-Fc. It was considered that each compound interferes at a concentration level enough to produce a variation of 10% in the initial response. According to this, the presence of epinephrine, ascorbic acid and dopamine in the sample interferes for concentrations of 5.0×10^{-6} M, 2.5×10^{-6} M, and 1.8×10^{-6} M, respectively. Note that the interference concentration level for all the compounds assayed ranges from 2 to 5 times higher than the VA-Fc concentration in solution.

Analytical application in real samples: determination of VA-Fc in serum

The applicability of the sensor was tested for VA-Fc determination in serum samples. As mentioned in the experimental section, human blood was centrifuged at 4 °C for 10 min at 2500 rpm. The serum was then separated with a syringe and stored at 4 °C prior to analysis. Samples were diluted 1:100 with phosphate buffer pH 7.0 and spiked with five different VA-Fc concentrations (from 1.4×10^{-6} M to 5.5×10^{-6} M). Dilution 1:100 allows obtaining a final concentration that both corresponds with possible physiological concentrations and fits in the linear range of the sensors. The VA-Fc determination was carried out by the DPV method (see experimental section). The recoveries, obtained for each VA-Fc concentration essayed, are summarized in Table 1. The recoveries were from 96% to 111%, indicating that the sensor can be successfully used to detect VA-Fc.

[VA-Fc] added (M)	[VA-Fc] found (M)	Recovery (%)
1.4×10^{-6}	1.6×10^{-6}	111
2.2×10^{-6}	2.3×10^{-6}	106
3.9×10^{-6}	4.1×10^{-6}	106
5.5×10^{-6}	5.3×10^{-6}	96

Table 1. Recoveries of VA-Fc from spiked serum samples

For evaluating the feasibility of the electrochemical method, serum samples were also analyzed by high-performance liquid chromatography (HPLC-UV) technique. Chromatographic conditions were selected according to reference 52 (see experimental section). Under these conditions, retention time of VA-Fc was 13.3 minutes. From the analysis of VA-Fc solutions of increasing concentrations, a good linearity of the calibration graph is obtained: $A = -1854 (\pm 7) + 179 (\pm 8) \times 10^7 C (M)$; $r = 0.999$. As described in experimental section, three different samples (spiked to obtain final VA-Fc concentrations of 1.0×10^{-5} , 2.5×10^{-5} and 5.0×10^{-5} M) were injected in the chromatographic system. Recoveries of 93%, 84% and 81%, respectively were obtained, demonstrating that similar recoveries are obtained by both methods.

4. Conclusions

In the present study, we have developed the first electrochemical sensor based on the combined employment of MoS₂ nanosheets and DNP as sensing elements and we have proved its applicability for determination of valproic acid previously derivatized with a ferrocene group. The best results were obtained for sensors developed employing MoS₂ dispersed in EtOH/water with a configuration consisting in a first layer of MoS₂ and a second layer of DNP. Concerning the morphology of the sensor surface, rather flat MoS₂ platelets are observed by AFM, being most of them almost totally covered by DNP. The presence of DNP significantly decreases the resistance charge transfer of the sensor surface compared to MoS₂. We prove that the combination of MoS₂ and DNP led to sensors with an enhanced analytical performance towards the oxidation of VA-Fc in comparison with sensors containing only one of the nanomaterials. Since an important synergistic effect takes place, the combination of these both emergent nanomaterials can offer new opportunities to develop electrochemical sensors with improved performances

for detection of different analytes, suggesting the possibility of a broad applicability. In our case, since valproic acid is not electroactive, its derivatization with a redox group is needed as previous step.

Acknowledgments

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Synergistic effect of MoS₂ and diamond nanoparticles in electrochemical sensors: determination of an anticonvulsant drug

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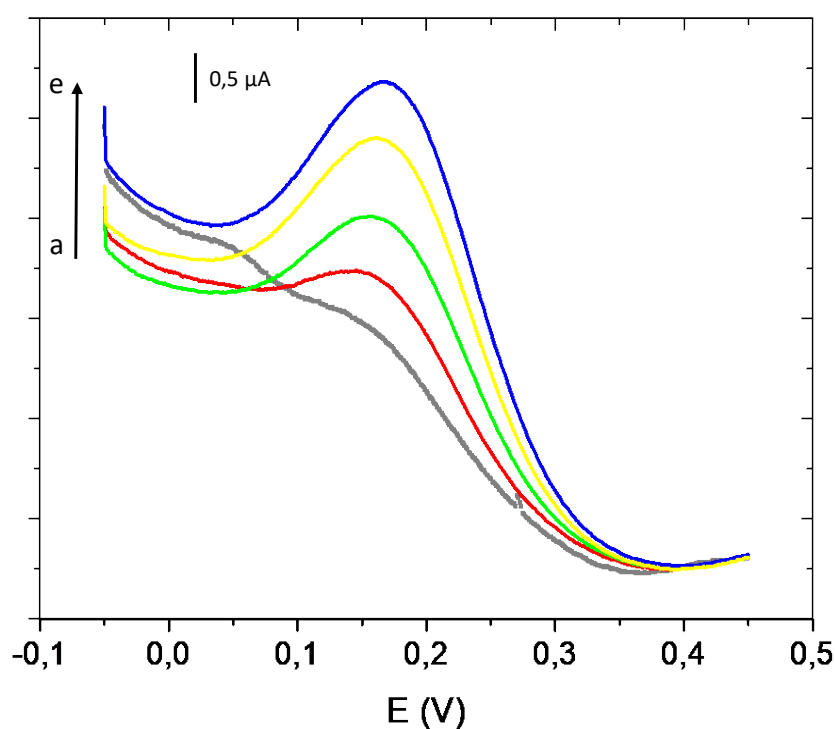


Figure S1. DPVs of GC/MoS₂/DNP sensor in a 0.1 M pH=7.0 phosphate buffer, containing increasing VA-Fc concentration: a) $5.5 \cdot 10^{-7}$ M, b) $1.4 \cdot 10^{-6}$ M, c) $2.2 \cdot 10^{-6}$ M, d) $3.0 \cdot 10^{-6}$ M and e) $3.9 \cdot 10^{-6}$ M. DPV conditions: range potential from -50 mV to 450 mV, scan rate 10 mV s^{-1} , pulse amplitude 60 mV and step potential 1 mV.

Table S1. Comparison of detection limits of valproic acid obtained by different methods

REFERENCE	METHOD / MATERIALS USED	DERIVATIZATION REAGENT	DETECTION LIMIT (µg/mL)
[35]	GC-MS	-----	1
[36]	LC-MS and CID-MS/MS	-----	0.31
[37]	RP-HPLC	Arylamide N-(2-hydroxyphenyl)-2-propylpentanamide	0.128
[38]	HPLC	2-bromo-2'-acetonaphthone	0.01
[39]	UPLC-MS/MS	2-picolylamine	0.03
[40]	HPLC-UV	2,4'-dibromoacetophenone	5
[41]	HPLC-MS/MS	4-dimethylaminobenzylamine dihydrochloride	0.2
[42]	HS-LPME-GC	-----	0.8
[43]	Capillary electrophoresis	-----	0.08
[44]	Electrochemical sensor (APTES-MNPs/PGE)	-----	0.4
[45]	Nanomechanical biosensor	-----	45
[46]	Proton sensor/Fluorescent detection (TGA-CdTe QDs)	-----	0.24
[47]	Potentiometric sensor- SIA (Mn(III)TPP-Cl/ MTES)	-----	130
Present work	Electrochemical sensor (GC/MoS ₂ /DNP)	Ferrocene	0.096

GC-MS (gas chromatography-mass spectrometry), LC-MS (liquid chromatography-mass spectrometry), CID-MS/MS (low-energy collision induced dissociation tandem mass spectrometry), RP-HPLC (reversed-phase high performance liquid chromatography), UPLC (ultraperformance liquid chromatography), HPLC-UV (high performance liquid chromatography-ultraviolet), HS-LPME-GC (headspace liquid phase microextraction-gas chromatography), APTES-MNPs (3-aminopropyletriethoxy silane coated magnetic nanoparticles), PGE (pencil graphite surface), TGA-CdTe QDs (thioglycolic acid -capped CdTe quantum dots), SIA (sequential-injection analysis system), Mn(III)TPP-Cl (Manganese (III) tetraphenylporphyrin), MTES (methyltriethoxysilane sol-gel), GC (Glassy carbon electrode), DNP (diamond nanoparticles).